



Static ML Project Audit Report

Adversarial examiner for machine learning notebooks



File analysed: credit-card-fraud-detection-using-machine-learning (3).ipynb

Audit type: Static notebook analysis with optional local AI examiner notes

Generated by: ModelHawk

Executive Summary

The notebook shows strong static audit signals. The detected project profile is Classification with Tabular data. ModelHawk detected positive evidence for evaluation quality, reproducibility and portfolio readiness. No major static red flags were detected, although the workflow should still be reviewed because static analysis cannot prove methodological correctness.

Important limitation. This report is based on static notebook analysis. It does not execute the notebook and does not prove the absence of data leakage, methodological issues or implementation errors. Instead, it highlights technical signals, likely risks and improvement opportunities detected from the notebook structure, code and markdown.

Report Scope and Methodology

Component	Description
Notebook parser	Reads code cells, markdown cells and notebook-level structure from the uploaded .ipynb file.
Static audit rules	Detects ML workflow signals such as splitting, metrics, reproducibility, resampling, pipelines, limitations and business context.
Heuristic scores	Converts detected signals into review dimensions for leakage risk, metric quality, reproducibility and portfolio readiness.
Local AI examiner	Optionally converts the deterministic audit result into clearer technical review notes using a local Ollama model.
PDF report	Compiles the structured audit into a professional LaTeX report suitable for portfolio review and technical discussion.

Table 1: ModelHawk report generation pipeline.

Detected Project Profile

Field	Detected value
Primary task	Classification
Secondary tasks	None detected
Data modality	Tabular
Problem traits	Imbalanced Data, Fraud Detection, Threshold Or Cost Sensitive, Deep Learning, Anomaly Or Rare Event Context
Applied audit packs	Universal, Classification, Tabular, Imbalanced Data, Deep Learning, Portfolio And Reporting
Confidence	High - 84/100

Table 2: Detected project profile and applied audit packs.

Notebook Structure

Metric	Value
Total cells	196
Code cells	80
Markdown cells	116

Table 3: Notebook structure summary.

Audit Scores

Dimension	Score	Interpretation
Leakage risk	8/100	Lower is better. This reflects static warning signals.
Metric quality	92/100	Higher suggests stronger metric coverage.
Reproducibility	90/100	Higher suggests better reproducibility signals.
Portfolio readiness	92/100	Higher suggests stronger presentation and documentation.

Table 4: ModelHawk static audit scores.

Score interpretation. Scores are heuristic static-analysis scores based on detected notebook signals. They are not statistical probabilities and do not guarantee correctness.

Audit Findings

No major static red flags detected

Severity: Low

The notebook passed the first static checks. ModelHawk detected several positive signals, although a deeper audit would require executing the workflow and testing robustness.

Good Practices Detected

The following positive signals were detected in the notebook. These do not prove correctness, but they indicate stronger ML project structure and documentation.

- Clear train/test split signal detected.
- Stratified split detected, useful for imbalanced classification.
- Pipeline or ColumnTransformer detected, which can reduce preprocessing leakage risk.

- Cross-validation or advanced validation signal detected.
- Random seed or random_state detected.
- PR-AUC or precision-recall evaluation detected.
- ROC-AUC evaluation detected.
- Precision, recall or F1-style evaluation detected.
- Threshold or probability-based decision logic detected.
- Hyperparameter search detected.
- Baseline comparison signal detected.
- Accuracy limitation appears to be acknowledged and supported by stronger metrics.
- Resampling/leakage awareness detected in the notebook explanation.
- Deep learning diagnostics such as early stopping or learning curves detected.
- Limitations or future work section detected.
- Business/cost context detected.
- Substantial markdown explanation detected.

Risks to Verify

These are not necessarily errors. They are review points that should be checked carefully before submission, publication or technical discussion.

- Verify that SMOTE/resampling is applied only to the training data in every experiment and cross-validation workflow.
- Accuracy is present, but stronger metrics and explanation were detected. Verify that the final conclusion prioritises minority-class performance.
- Metric coverage looks strong statically, but verify that the metrics are calculated on held-out data only.

Priority Fixes

- No urgent priority fixes were returned by the current audit.

Recommendations

1. State the project type, target variable and evaluation objective clearly.
2. Make the final metric choice explicit and explain which metric should drive model selection.
3. Strengthen the final discussion by clearly comparing the selected model against the baseline.
4. Keep the resampling/leakage explanation close to the validation methodology.
5. Make dependency versions and rerun instructions easy to find.

6. Connect the limitations section more directly to deployment risk and future validation.
7. Strengthen the business/cost interpretation of false positives and false negatives.

Suggested Notebook Improvements

The following markdown suggestions are designed to improve the explanation and defensibility of the notebook.

Final metric justification

Where: Before the final model selection or conclusion

Although accuracy is reported, the final model should be selected using metrics that better reflect minority-class performance. In this project, precision, recall, F1-score and PR-AUC provide a more informative view of fraud detection performance than accuracy alone.

Local AI Examiner Notes

No local AI examiner narrative was included in this PDF request. The deterministic static audit sections above were still generated successfully.

Viva and Interview Preparation

These questions are designed to help the author defend modelling choices, evaluation strategy and project limitations in a technical discussion.

1. Why is accuracy not enough for this project?
2. How did you make sure resampling did not create leakage?
3. How did you choose your decision threshold?
4. Which metric would you prioritise if precision and recall move in opposite directions?
5. How would you test whether this model generalises to new data?
6. What are the main limitations of this project?

Detected Technical Signals

The following static signals were detected from the notebook code and markdown.

- Acknowledges Accuracy Limitation
- Discusses Resampling Safety
- Has Markdown Explanation
- Mentions Business Context

- Mentions Conclusion
- Mentions Imbalance
- Mentions Limitations
- Uses Accuracy
- Uses Anomaly Signals
- Uses Baseline
- Uses Classification Model
- Uses Classification Report
- Uses Confusion Matrix
- Uses Cross Validation
- Uses Deep Learning Signals
- Uses Early Stopping
- Uses Hyperparameter Search
- Uses Learning Curves
- Uses Multiple Models
- Uses Pipeline
- Uses Prauc
- Uses Precision Recall F1
- Uses Random State
- Uses Requirements Or Dependencies
- Uses Roc Auc
- Uses Smote
- Uses Stratify
- Uses Threshold Tuning
- Uses Train Test Split
- Uses Train Validation Split

Final Note

Generated by ModelHawk. This automatically generated report should be interpreted as technical guidance, not as a final guarantee of correctness. A complete review would require executing the notebook, validating the data pipeline, checking runtime outputs and reviewing the modelling decisions in context.